Name : Gaurang Tyagi

Roll No : 16

Cyber Security Assignment - 1

Task 1 : Sending an encrypted email

A. Practical Implementation

> Student A's (Gaurang Tyagi) public key


```
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Student B's (Ravish Ranjan) public key


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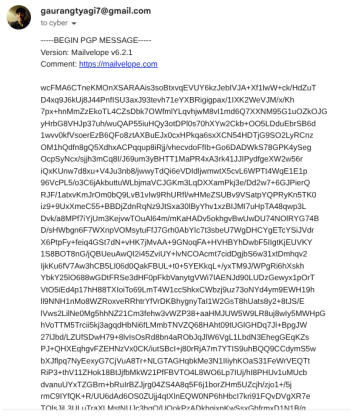
Mail Recieved by Student B (Ravish Ranjan)

Raw email



gaurangtyagi7@gmail.com
to cyber

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Version: Mailvelope v6.2.1
Comment: https://mailvelope.com

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Decrypted Email

Dear Ravish,

I hope you are doing well.

Please find below the details of our agreed Chemical Supply Agreement for the supply of sodium bicarbonate:

Agreement Date: 18 April 2026
Supplier: Gaurang Tyagi
Buyer: Ravish Ranjan

Product Details:
Chemical: Sodium Bicarbonate

Grade: Technical Grade

Quantity: 1 Metric Ton (1000 kg)

Delivery Details:
Location: Department of Computer Science, Faculty of Mathematical Sciences, University of Delhi

Schedule: Within 7-10 working days from purchase order confirmation (single lot)

Mode of Transport: Road transport via covered truck

Pricing:
Unit Price: ₹22 per kg

Total Contract Value: ₹25,960 (including 18% GST)

Payment Terms:
Net 30 days from the date of invoice

Additional Terms:
Inspection period: 30 days from delivery

Governing Law: Laws of India, jurisdiction in Delhi

Dispute Resolution: Negotiation followed by arbitration/legal proceedings in Delhi

Kindly review the above details and confirm your acceptance so we can proceed further.

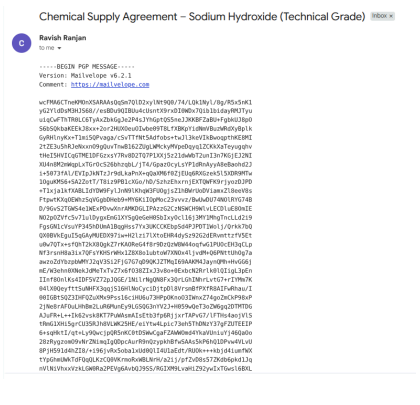
Looking forward to your confirmation.

Best regards,
Gaurang Tyagi
Department of Computer Science
University of Delhi

Signed | Gaurang Tyagi <gaurangtyagi7@gmail.com> 8D25BAFC8D8DB4D4f6

Mail Recieved by Student A (Gaurang Tyagi)

Raw email



Chemical Supply Agreement – Sodium Hydroxide (Technical Grade) Inbox

Ravish Ranjan
to me

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Comment: https://mailvelope.com

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Decrypted Email

Dear Gaurang,

I hope you are doing well.

Please find below the details of our agreed Chemical Supply Agreement for the supply of sodium hydroxide:

Agreement Date: 18 April 2026
Supplier: Ravish Ranjan
Buyer: Gaurang Tyagi

Product Details:
Chemical: Sodium Hydroxide

Grade: Technical Grade

Quantity: 1 Metric Ton (1000 kg)

Delivery Details:
Location: Department of Computer Science, Faculty of Mathematical Sciences, University of Delhi

Schedule: Within 7-10 working days from purchase order confirmation (single lot)

Mode of Transport: Road transport via covered truck

Pricing:
Unit Price: ₹15 per kg

Total Contract Value: ₹17,700 (including 18% GST)

Payment Terms:
Net 30 days from the date of invoice

Additional Terms:
Inspection period: 30 days from delivery

Governing Law: Laws of India, jurisdiction in Delhi

Dispute Resolution: Negotiation followed by arbitration/legal proceedings in Delhi

Kindly review the above details and confirm your acceptance so we can proceed further.

Looking forward to your confirmation.

Best regards,
Ravish Ranjan
Department of Computer Science
University of Delhi

Signed | cyber security assignment <ravishranjan2003@gmail.com> 8BBF5D2587E0AC8E

B. Indian Cyber Law and Compliance

Company liability :

> Under Section 43A of the IT Act, 2000, can the company be legally penalised for failing to protect this sensitive information?

```
Yes, the company will be legally penalised for failing to protect this
sensitive information as
under Section 43A, companies are liable to pay compensation if negligent in
maintaining reasonable security practices, leading to wrongful loss or gain.
This applies to "body corporates" handling sensitive personal data, such as
financial or health information, with no specific upper limit on
compensation.
```

Mandatory Reporting :

> According to CERT-in rules, what are the steps the company must take to report this cyber incident to the government ?

```
Steps that a company can take to report a cyber incident to the government
are :

1. Identify the Incident Type: Determine if the event falls under the 20
categories of mandatory reportable incidents listed in Annexure I, such as
data breaches, ransomware, unauthorized access, or attacks on critical
infrastructure.

2. Report within 6 Hours: The incident must be reported to CERT-In within 6
hours of noticing it or being informed about it. The clock starts the moment
the IT team, security vendor, or any employee detects the incident, not when
the investigation concludes.

3. Use Authorized Reporting Channels: Report the incident via:
    - Email: incident@cert-in.org.in
    - Website/Portal: https://www.cert-in.org.in
    - Phone Hotline: 1800-11-4949

4. Submit information in required format: Provide the report according to
the formats published on the CERT-In website. The report should contain, as
much as possible:
    - Time of occurrence
    - Information regarding affected systems/network
    - Symptoms observed
    - Technical information (security systems, actions taken)
    - Incident impact and type

5. Designate a Point of Contact (PoC): Entities must designate a PoC to
interface with CERT-In and send their details (Name, Address, Email, Phone)
```

```
to info@cert-in.org.in as per Annexure II.

6. Maintain and Provide Logs: Enable logs for all ICT systems and maintain
them securely within Indian jurisdiction for a rolling period of 180 days.
These must be provided to CERT-In upon reporting an incident or when
requested.

7. Ensure Time Synchronization: All ICT system clocks must be synchronized
with the Network Time Protocol (NTP) Server of the National Informatics
Centre (NIC) or National Physical Laboratory (NPL).

Consequences of Non-Compliance:
Failure to report within the 6-hour window or non-compliance with these
directions can lead to penalties under Section 70B(7) of the IT Act, 2000,
which includes imprisonment for up to one year or a fine of up to ₹1 lakh,
or both.
```

Task 2: Website Cloning, Phishing and Credential Harvesting using SET

A. Clone the target website

> Step 1 : Launch SET and navigate to the Credential Harvester

```
  1) Java Applet Attack Method
  2) Metasploit Browser Exploit Method
  3) Credential Harvester Attack Method
  4) Tabnabbing Attack Method
  5) Web Jacking Attack Method
  6) Multi-Attack Web Method
  7) HTA Attack Method

 99) Return to Main Menu

set:webattack>3
```

> Step 2 : Enter the IP address of the POST-back server

```
set:webattack> IP address for the POST back in Harvester/Tabnabbing [192.168.1.4]:
[-] SET supports both HTTP and HTTPS
[-] Example: http://www.thisisafakesite.com
set:webattack> Enter the url to clone: http://hackthissite.org
```

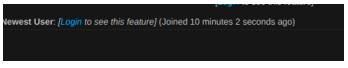
> Step 3 : Enter the target URL to clone

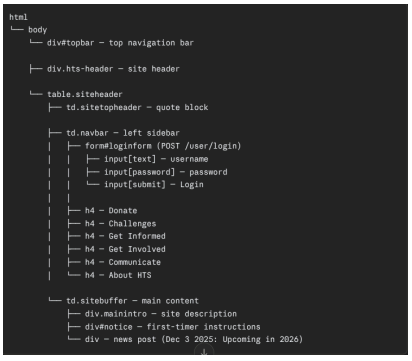
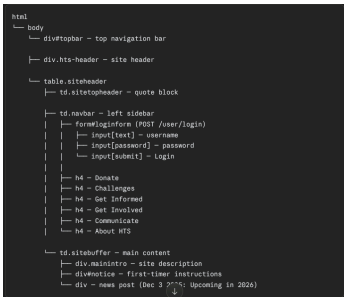
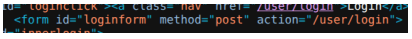
```
set:webattack> IP address for the POST back in Harvester/Tabnabbing [192.168.1.4]:
[-] SET supports both HTTP and HTTPS
[-] Example: http://www.thisisafakesite.com
set:webattack> Enter the url to clone: http://hackthissite.org
```

Step 4 : Access the cloned site

Original Site	Cloned Site

B. Comparative Analysis - Original vs Cloned Website

Comparison Criteria	Original Website	Cloned Website	Difference / Scurity Implications
URL Domain	https://hackthissite.org	http://192.168.1.4	
SSL/TLS Certificate Status	present	not present	
Page Visual Appearance			In the original website newest user time is

Comparison Criteria	Original Website	Cloned Website	Difference / Scurity Implications
			updating dynamically whereas in the cloned website it is static which is a sign that the site is cloned.
HTML Source Code Structure			In cloned site the anchor tags all point to the original site
Form Action Attribute			 because of this tag the action attribute of cloned site's form takes user to the actual site when login form is submitted
Javascript External Links	JQuery link	JQuery Link	There is no difference
Image and resource loading	Normal Image and resource loading	Normal Image and resource loading	There is no difference
Login form behaviour	In the original website the login form works normally , original website does not track	In the cloned website the login form redirects the user to the actual	This behaviour can cause the user to think that the cloned

Comparison Criteria	Original Website	Cloned Website	Difference / Scurity Implications
	whether the user clicked on password field or username field.	hackthissite.org website, cloned website tracks whether the user clicked on password field or username field	website is the legitimate website.

> One paragraph explaining how a vigilant user could identify the fake site

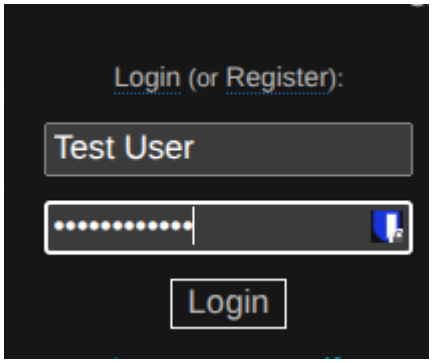
```
A vigilant user can identify a fake website by carefully examining several
indicators. First, they should check the URL in the address bar to ensure it
exactly matches the legitimate domain, as fake sites often use slight
variations like in this case it is hosted on 192.168.1.4 . They should also
verify the presence of HTTPS and inspect the security certificate for
authenticity like in this case the cloned website does not use HTTPs
protocol. Inconsistencies in design, broken links, or outdated content can
also signal a cloned or fake page. Additionally, users should be cautious if
the page behaves unusually, such as redirecting unexpectedly after login.
Using browser tools to inspect page details and avoiding clicking on
suspicious links further helps in detecting fraudulent websites.
```

> Explain what browser indicator (URL bar, padlock, certificate details) is most reliable for detecting a phishing site

```
URL bar is most reliable for detecting a phishing site as most malicious
sites use visually similar but incorrect domain names to trick users, and
the URL directly reveals the actual domain being accessed.
```

C. Credential Harvesting Demonstration

> Step 1 : Submit test credentials on the cloned site.



> Step 2 : Observe captured data in SET

```
192.168.1.4 - - [19/Apr/2026 03:04:10] "GET /images/favicon.ico HTTP/1.1" 404 -
[*] WE GOT A HIT! Printing the output:
POSSIBLE USERNAME FIELD FOUND: login=Test+User
POSSIBLE PASSWORD FIELD FOUND: password=testpassword
PARAM: security_level=0
PARAM: form=submit
[*] WHEN YOU'RE FINISHED, HIT CONTROL-C TO GENERATE A REPORT.
```

> Step 3 : Review the SET harvest log file.

```
<?xml version="1.0" encoding='UTF-8'?>
<harvester>
   URL=http://hackthissite.org
   <url>      <param>login=Test+User</param>
      <param>password=testpassword</param>
      <param>security_level=0</param>
      <param>form=submit</param>
   </url>
</harvester>
```

> Describe the complete victim experience from click to redirect — what would the victim see and believe?

```
1. When a user clicks a link leading to a cloned webpage, they are presented
with a page that visually appears identical to the legitimate website,
including branding, layout, and login form.
2. Believing it to be authentic, the user enters their credentials.
3. After submitting the form, the page redirects them to the actual website
, making the interaction seem successful and legitimate.
4. Since the final page appears genuine and no immediate error is shown, the
user is unlikely to suspect that anything unusual has occurred and may
believe the login process worked normally.
```

> If this were a real attack against a banking site, what is the worst-case impact of a successful credential harvest?

```
If this were a real attack against a baking site  then successful credential
harvesting can have severe consequences. An attacker gaining access to valid
login credentials could:

1. Access the user’s bank account
2. Perform unauthorized transactions or transfers
3. Steal sensitive financial and personal information
4. Change account details (email, phone number) to lock out the user
5. Initiate further fraud such as identity theft or loan applications
In the worst case, this can lead to significant financial loss, privacy
compromise.
```

D. Detection, Prevention and Incident Response

> What are the three visual indicators in the browser that could help a user identify a cloned phising site?

```
1. Incorrect URL: for example misspelled domains like g00gle.com or
amaz0n.com
2. Padlock: This indicates whether the connection is secure or not. Abscence
of this is suspicious.
3. Certificate Details: If the padlock is present then a user can click on
it to check who issued the certificate and the domain name it is issued to,
if this information does not match the expected organization then it is a
phising site.
```

> How does SET's Credential Harvester avoid detection by the victim during and after the attack?

```
SET's Credential Harvester avoid detection by the victim in the following
ways ->

1. Looks identical to the real site: The page copies branding, layout, and
the login form, so nothing appears out of place to a quick glance.
2. Uses normal web flows: Submitting a login form and then seeing a familiar
page  after a redirect feels like a routine login experience.
3. No immediate error signals: The page often doesn’t show obvious warnings
or failures, so the user has no clear reason to suspect anything.
4. Time gap between cause and effect: Any consequences (e.g. account misuse)
happen later, making it hard to link them to that login event.
```

> How would implementing HTTPS with a valid TLS certificate on the cloned site make detection significantly harder for a typical user?

```
Implementing HTTPS with a valid TLS certificate on a cloned site makes
detection harder because it creates a false sense of security for typical
users. When users see the padlock icon and “https://” in the address bar,
they often assume the website is trustworthy. However, HTTPS only ensures
that the connection between the user and the website is encrypted, not that
the website itself is legitimate. Attackers can obtain valid TLS
certificates for domains they control, so a phishing site can appear secure
even though it is malicious. As a result, users who rely solely on the
padlock indicator may fail to notice subtle differences in the domain name
or other warning signs, making the cloned site more convincing and difficult
to identify.
```

> What network-level controls (DNS filtering, proxy inspection, SIEM alerts) could detect and block a credential harvesting attack before credentials are submitted?

```
Network-level controls like
- DNS filtering can prevent users from reaching a malicious pages and detect
misspelled domain names.
- A web proxy can inspect and control outbound web traffic in real time and
perform TLS inspection to analyze HTTPs contents.
- SIEM alerts (Security Information and Event Management alerts.) can alert
on access to newly seen domains, multiple users visiting the same suspicious
URL, login POST requests to non-corporate domains and integrate with threat
intelligence feeds
This Enables early detection and rapid response before widespread impact.
```

> Compare the effectiveness of technical controls vs user security awareness training in defending against phishing attacks. Which provides more reliable long-term protection and why?

Technical Controls	User Security Awareness
It is automated and consistent in defending against attacks	It is not reliable as humans are prone to making error
It is highly scalable	It is not highly scalable
It needs constant updates to be able to catch new attacks	It improves with training

Technical controls provide more reliable and long-term protection because :

Technical controls such as DNS filtering, secure web gateways, email filtering, and multi-factor authentication provide automated protection. They work continuously, do not rely on human judgment, and can block malicious websites or emails before the user interacts with them. This makes them highly reliable and scalable across an organization. However, they may fail to detect new or sophisticated phishing attacks.

On the other hand, user security awareness training helps individuals recognize phishing attempts and suspicious behavior. It is effective against social engineering techniques and helps build a security-conscious culture. However, it is less reliable because it depends on human behavior. Users may make mistakes, especially under pressure, and the effectiveness of training may decrease over time without regular reinforcement.
In comparison, technical controls provide more consistent and reliable long-term protection because they operate automatically and reduce the chances of human error